REFINED QUALITY ASSURANCE TESTING DOCUMENTATION (QATD)

Team Grabbeez

UMHackathon 2026

Document Control

Field	Detail
System Under Test (SUT)	Stocky AI
Team Repo URL	https://github.com/Vaarshh17/StockyAI
Project Board URL	-
Live Deployment URL	https://stocky-ai-dashboard.vercel.app
Objective	Ensure Stocky AI accurately logs inventory and trades, triggers proactive scheduler jobs (Morning Brief, Spoilage, Velocity, Credit) flawlessly, and handles LLM tool-calling loops without hallucinating or looping infinitely.

FINAL ROUND (Execution, RCA & Hard Evidence)

1. Test Execution & Evidence (Manual)

The following manual tests map directly to the demo scenarios developed in the `tests/scenarios/` directory.

Test Case ID	Test Type & Mapped Feature	Test Description	Expected Result	Actual Result	Status	Proof of Execution
TC-01	Happy Case: Proactive Morning Brief	Verifies the 3:30 AM daily scheduler job compiles	System successfully fetches db queries and generates	Brief successfully generated. Output was over	PASS	scheduled_morning_brief test output

Test Case ID	Test Type & Mapped Feature	Test Description	Expected Result	Actual Result	Status	Proof of Execution
		inventory, weather, and credit data into a single briefing.	a response > 100 characters containing stock, credit, and weather.	100 characters and included all required elements.		
TC-02	Happy Case: Inventory Update + Velocity	User logs a delivery via natural language ("Dapat 2 tan tomato dari Pak Ali, RM2.60 sekilo").	LLM identifies the `update_inventory` tool, parses 2000kg, and updates the database.	Bot responded confirming "tomato" and "2000" kg.	PASS	scheduled_inventory test output
TC-03	Happy Case: Credit Ledger	User logs a credit sale ("Kedai Pak Hamid ambil 500kg bayam, bayar Jumaat, RM800") and queries outstanding credit.	`log_credit` tool is called. Subsequent query calls `get_outstanding_credit` and returns Hamid's debt.	Bot successfully identified "hamid" and the "800" RM balance.	PASS	credit_ledger test output

Test Case ID	Test Type & Mapped Feature	Test Description	Expected Result	Actual Result	Status	Proof of Execution
TC-AI-01	AI Output Validation: Draft Approvals	Sent a prompt asking the bot to draft a message to a supplier (e.g., "Draftkan mesej untuk Pak Ali").	The agent detects `DRAFT_MESSAGE::` format, flags `needs_approval=True`, and returns an inline keyboard.	Agent correctly flagged the message for approval and generated a UUID draft_id.	PASS	agent_approval_require output image

---

2. Automated Testing Pipeline

2.1. Unit Testing (Core Logic & Functions)

Unit testing focused on the isolation of the database tool functions mapped in `tests/test_tools.py`.

- **Test Runner Used:** `pytest (python)` and `pytest-asyncio`
- **Targeted Components:** Database Queries (`db/queries.py`)
- **File(s) Covered:** `tests/test_tools.py`

Unit Test ID	Function/Component Tested	Test Scenario	Expected Outcome	Actual Result	Status
UT-01	`db_update_inventory()`	Insert 500kg of tomato from Pak Ali and check if the total stock sum	`after_qty == before_qty + 500`.	`after_qty` correctly reflected the +500 increment.	PASS

Unit Test ID	Function/Component Tested	Test Scenario	Expected Outcome	Actual Result	Status
		increases exactly by 500.			
UT-02	`db_get_velocity()`	Verify the velocity calculation computes the average daily kg sold accurately over 7 days.	Returns a dictionary with an `avg_daily_kg` > 0.	Returned correctly structured dict with `avg_daily_kg` properly computed.	PASS
UT-03	`db_compare_prices()`	Fetch a ranked list of supplier prices for a commodity compared to the FAMA benchmark.	Result contains `suppliers` and `fama_benchmark` keys, sorted ascending by `price_per_kg`.	List returned perfectly sorted by price.	PASS
UT-04	`transcribe_voice()`	Test how the Whisper voice transcription handles empty bytes, `None` inputs, and simulated model failures.	Returns an empty string `""` without crashing the application.	Safely caught exceptions and returned empty strings.	PASS

Unit Test ID	Function/Component Tested	Test Scenario	Expected Outcome	Actual Result	Status
UT-05	`get_instinct()`	Verify that Stocky's Instinct successfully parses mocked data from inventory, digests, and price trends, and prepends the "Stocky nampak sesuatu:" prefix.	String returned starts with the correct prefix and includes the insights.	Prefix appended correctly; handled empty LLM responses by returning "semua ok".	PASS
UT-06	`get_forecast()`	Mock the `httpx.AsyncClient` to simulate fetching the Open-Meteo daily forecast for Kuala Lumpur.	Returns a list of dictionaries containing keys like `condition` and `is_rainy`.	Correctly parsed weather codes (e.g., 0 to "Cerah") and returned the forecast list.	PASS

UT-01

```
═══════════════════════════════════════════════════════ 16 passed in 0.11s ═══════════════════════════════════════════════════

StockyAI on  main [!+?] via  v3.11.11 (.venv)
❯ python -m pytest tests/test_tools.py -v -k "test_update_inventory"

═══════════════════════════════ test session starts ═══════════════════════════════
platform darwin -- Python 3.11.11, pytest-9.0.3, pluggy-1.6.0 -- /Users/weijie/code/StockyAI/.venv/bin/python
cachedir: .pytest_cache
rootdir: /Users/weijie/code/StockyAI
configfile: pytest.ini
plugins: cov-7.1.0, asyncio-1.3.0, anyio-4.13.0
asyncio: mode=Mode.STRICT, debug=False, asyncio_default_fixture_loop_scope=None, asyncio_default_test_loop_scope=function
collected 16 items / 15 deselected / 1 selected

tests/test_tools.py::TestExecuteTool::test_update_inventory PASSED

═══════════════════ 1 passed, 15 deselected in 0.10s ═══════════════════
```

UT-02

```
StockyAI on  main [!+?] via  v3.11.11 (.venv)
❯ python -m pytest tests/test_tools.py -v -k "test_get_velocity"

═══════════════════════════════ test session starts ═══════════════════════════════
platform darwin -- Python 3.11.11, pytest-9.0.3, pluggy-1.6.0 -- /Users/weijie/code/StockyAI/.venv/bin/python
cachedir: .pytest_cache
rootdir: /Users/weijie/code/StockyAI
configfile: pytest.ini
plugins: cov-7.1.0, asyncio-1.3.0, anyio-4.13.0
asyncio: mode=Mode.STRICT, debug=False, asyncio_default_fixture_loop_scope=None, asyncio_default_test_loop_scope=function
collected 16 items / 15 deselected / 1 selected

tests/test_tools.py::TestExecuteTool::test_get_velocity PASSED

═══════════════════ 1 passed, 15 deselected in 0.10s ═══════════════════
```

UT-03

```
StockyAI on  main [!+?] via  v3.11.11 (.venv)
❯ python -m pytest tests/test_tools.py -v -k "test_compare_supplier_prices"

═══════════════════════════════ test session starts ═══════════════════════════════
platform darwin -- Python 3.11.11, pytest-9.0.3, pluggy-1.6.0 -- /Users/weijie/code/StockyAI/.venv/bin/python
cachedir: .pytest_cache
rootdir: /Users/weijie/code/StockyAI
configfile: pytest.ini
plugins: cov-7.1.0, asyncio-1.3.0, anyio-4.13.0
asyncio: mode=Mode.STRICT, debug=False, asyncio_default_fixture_loop_scope=None, asyncio_default_test_loop_scope=function
collected 16 items / 15 deselected / 1 selected

tests/test_tools.py::TestExecuteTool::test_compare_supplier_prices PASSED

═══════════════════ 1 passed, 15 deselected in 0.10s ═══════════════════
```

UT-04

```
StockyAI on main [!+?] via v3.11.11 (.venv) took 4m39s
❯ python -m pytest tests/test_voice.py -v

============================================================================ test session starts ============================================================================
platform darwin -- Python 3.11.11, pytest-9.0.3, pluggy-1.6.0 -- /Users/weijie/code/StockyAI/.venv/bin/python
cachedir: .pytest_cache
rootdir: /Users/weijie/code/StockyAI
configfile: pytest.ini
plugins: cov-7.1.0, asyncio-1.3.0, anyio-4.13.0
asyncio: mode=Mode.STRICT, debug=False, asyncio_default_fixture_loop_scope=None, asyncio_default_test_loop_scope=function
collected 6 items

tests/test_voice.py::TestTranscribeVoice::test_empty_bytes_returns_empty PASSED
tests/test_voice.py::TestTranscribeVoice::test_none_bytes_returns_empty PASSED
tests/test_voice.py::TestTranscribeVoice::test_model_failure_returns_empty PASSED
tests/test_voice.py::TestTranscribeVoice::test_successful_transcription PASSED
tests/test_voice.py::TestTranscribeVoice::test_language_hint_passed PASSED
tests/test_voice.py::TestTranscribeVoice::test_transcription_failure_returns_empty PASSED

============================================================================= 6 passed in 0.03s =============================================================================
```

UT-05

```
StockyAI on main [!+?] via v3.11.11 (.venv)
❯ python -m pytest tests/test_tools.py -v -k "test_get_instinct_analysis"

============================================================================ test session starts ============================================================================
platform darwin -- Python 3.11.11, pytest-9.0.3, pluggy-1.6.0 -- /Users/weijie/code/StockyAI/.venv/bin/python
cachedir: .pytest_cache
rootdir: /Users/weijie/code/StockyAI
configfile: pytest.ini
plugins: cov-7.1.0, asyncio-1.3.0, anyio-4.13.0
asyncio: mode=Mode.STRICT, debug=False, asyncio_default_fixture_loop_scope=None, asyncio_default_test_loop_scope=function
collected 16 items / 15 deselected / 1 selected

tests/test_tools.py::TestExecuteTool::test_get_instinct_analysis PASSED

============================================================================ 1 passed, 15 deselected in 0.10s ============================================================================
```

UT-06

```
StockyAI on main [!+?] via v3.11.11 (.venv)
❯ python -m pytest tests/test_tools.py -v -k "test_get_weather_forecast"

============================================================================ test session starts ============================================================================
platform darwin -- Python 3.11.11, pytest-9.0.3, pluggy-1.6.0 -- /Users/weijie/code/StockyAI/.venv/bin/python
cachedir: .pytest_cache
rootdir: /Users/weijie/code/StockyAI
configfile: pytest.ini
plugins: cov-7.1.0, asyncio-1.3.0, anyio-4.13.0
asyncio: mode=Mode.STRICT, debug=False, asyncio_default_fixture_loop_scope=None, asyncio_default_test_loop_scope=function
collected 16 items / 15 deselected / 1 selected

tests/test_tools.py::TestExecuteTool::test_get_weather_forecast PASSED

============================================================================ 1 passed, 15 deselected in 0.09s ============================================================================
```

2.2. Integration & API Testing (System Modules)

Integration testing focused on the core Agent Loop connecting the User, the LLM (`services/glm.py`), and the Local Database.

- **Test Tool Used:** `pytest(python), pytest-asyncio` and `unittest.mock`
- **Targeted Integrations:** Database (SQLite/SQLAlchemy), Scheduler (APScheduler), and Bot Handlers (python-telegram-bot)
- **File(s) Covered:** tests/test_agent.py, `tests/integration/test_db.py`, `tests/integration/test_scheduler.py`, `tests/integration/test_bot.py`

Test ID	Integration / Point Tested	Test Scenario	Expected Outcome	Actual Result	Status
IT-01	Database Models & Queries	Verify that `init_db()` successfully creates all SQLAlchemy tables and that functions like `db_update_inventory` properly execute SQL inserts.	Tables (inventory, suppliers, trades, etc.) are created in the SQLite in-memory DB; data is committed and retrievable.	Tables confirmed created; inventory updates successfully cascaded to trade logs.	PASS
IT-02	Proactive Scheduler Jobs	Test that `start_scheduler()` successfully registers all 5 cron jobs and that jobs like `spoilage_check_job` only message users when	Scheduler registers exactly 5 jobs; spoilage job triggers `send_message` only when `days_remaining <= 2`.	Jobs registered correctly; alerts fired accurately based on mocked db data.	PASS

Test ID	Integration / Point Tested	Test Scenario	Expected Outcome	Actual Result	Status
		risk thresholds are met.			
IT-03	Telegram Bot Handlers	Pass simulated Telegram `Update` and `Context` objects to `handle_message` to test message routing, voice transcription, and onboarding gates.	Messages are routed correctly (text vs. voice vs. forwarded) and onboarding states update DB appropriately.	Handlers accurately parsed mocked messages and triggered the agent loop.	PASS

IT-01

```
StockyAI on ⎇ main [!+?] via 🐍 v3.11.11 (.venv) took 1m40s
❯ python -m pytest tests/integration/test_db.py -v

============================================================ test session starts ============================================================
platform darwin -- Python 3.11.11, pytest-9.0.3, pluggy-1.6.0 -- /Users/weijie/code/StockyAI/.venv/bin/python
cachedir: .pytest_cache
rootdir: /Users/weijie/code/StockyAI
configfile: pytest.ini
plugins: cov-7.1.0, asyncio-1.3.0, anyio-4.13.0
asyncio: mode=Mode.STRICT, debug=False, asyncio_default_fixture_loop_scope=None, asyncio_default_test_loop_scope=function
collected 30 items

tests/integration/test_db.py::test_init_db_creates_tables PASSED
tests/integration/test_db.py::TestInventoryQueries::test_get_inventory_empty_db PASSED
tests/integration/test_db.py::TestInventoryQueries::test_update_inventory_creates_entry PASSED
tests/integration/test_db.py::TestInventoryQueries::test_update_inventory_creates_supplier PASSED
tests/integration/test_db.py::TestInventoryQueries::test_update_inventory_logs_trade PASSED
tests/integration/test_db.py::TestInventoryQueries::test_update_inventory_without_supplier PASSED
tests/integration/test_db.py::TestInventoryQueries::test_log_sell_deducts_fifo PASSED
tests/integration/test_db.py::TestInventoryQueries::test_log_sell_records_trade PASSED
tests/integration/test_db.py::TestInventoryQueries::test_days_remaining_calculation PASSED
tests/integration/test_db.py::TestInventoryQueries::test_zero_quantity_excluded PASSED
tests/integration/test_db.py::TestCreditQueries::test_log_and_get_credit PASSED
tests/integration/test_db.py::TestCreditQueries::test_get_all_unpaid_credit PASSED
tests/integration/test_db.py::TestCreditQueries::test_overdue_detection PASSED
tests/integration/test_db.py::TestCreditQueries::test_due_today_detection PASSED
tests/integration/test_db.py::TestCreditQueries::test_log_credit_default_due_date PASSED
tests/integration/test_db.py::TestSupplierPriceQueries::test_compare_prices PASSED
tests/integration/test_db.py::TestSupplierPriceQueries::test_compare_prices_empty PASSED
tests/integration/test_db.py::TestVelocityQueries::test_velocity_calculation PASSED
tests/integration/test_db.py::TestVelocityQueries::test_velocity_no_sales PASSED
tests/integration/test_db.py::TestFamaBenchmarkQueries::test_get_fama_price PASSED
tests/integration/test_db.py::TestFamaBenchmarkQueries::test_get_fama_price_nonexistent PASSED
tests/integration/test_db.py::TestWeeklyDigest::test_digest_with_data PASSED
tests/integration/test_db.py::TestWeeklyDigest::test_digest_empty_db PASSED
tests/integration/test_db.py::TestUserProfile::test_save_and_get_persona PASSED
tests/integration/test_db.py::TestUserProfile::test_update_existing_persona PASSED
tests/integration/test_db.py::TestUserProfile::test_get_nonexistent_persona PASSED
tests/integration/test_db.py::TestPriceTrend::test_price_trend_with_data PASSED
tests/integration/test_db.py::TestPriceTrend::test_price_trend_no_data PASSED
tests/integration/test_db.py::TestSeedData::test_seed_creates_data PASSED
tests/integration/test_db.py::TestSeedData::test_seed_idempotent PASSED

============================================================= 30 passed in 0.66s ============================================================
```

IT-02

```
StockyAI on main [!+?] via v3.11.11 (.venv)
❯ python -m pytest tests/integration/test_scheduler.py -v

============================================================ test session starts ============================================================
platform darwin -- Python 3.11.11, pytest-9.0.3, pluggy-1.6.0 -- /Users/weijie/code/StockyAI/.venv/bin/python
cachedir: .pytest_cache
rootdir: /Users/weijie/code/StockyAI
configfile: pytest.ini
plugins: cov-7.1.0, asyncio-1.3.0, anyio-4.13.0
asyncio: mode=Mode.STRICT, debug=False, asyncio_default_fixture_loop_scope=None, asyncio_default_test_loop_scope=function
collected 13 items

tests/integration/test_scheduler.py::TestStartScheduler::test_start_scheduler_registers_jobs PASSED
tests/integration/test_scheduler.py::TestStartScheduler::test_start_scheduler_replaces_existing PASSED
tests/integration/test_scheduler.py::TestMorningBriefJob::test_skips_if_no_active_users PASSED
tests/integration/test_scheduler.py::TestMorningBriefJob::test_sends_brief_to_active_users PASSED
tests/integration/test_scheduler.py::TestSpoilageCheckJob::test_no_alert_when_no_risk PASSED
tests/integration/test_scheduler.py::TestSpoilageCheckJob::test_alerts_when_spoilage_risk PASSED
tests/integration/test_scheduler.py::TestVelocityAlertJob::test_no_alert_when_normal PASSED
tests/integration/test_scheduler.py::TestVelocityAlertJob::test_alerts_on_low_stock PASSED
tests/integration/test_scheduler.py::TestCreditReminderJob::test_no_alert_when_no_urgent_credit PASSED
tests/integration/test_scheduler.py::TestCreditReminderJob::test_alerts_on_overdue_credit PASSED
tests/integration/test_scheduler.py::TestMondayDigestJob::test_sends_digest PASSED
tests/integration/test_scheduler.py::TestSendToAllUsers::test_sends_to_all_active_users PASSED
tests/integration/test_scheduler.py::TestSendToAllUsers::test_handles_send_failure_gracefully PASSED

============================================================= 13 passed in 0.66s ============================================================
```

IT-03

```
StockyAI on main [!+?] via v3.11.11 (.venv)
❯ python -m pytest tests/integration/test_bot.py -v

============================================================ test session starts ============================================================
platform darwin -- Python 3.11.11, pytest-9.0.3, pluggy-1.6.0 -- /Users/weijie/code/StockyAI/.venv/bin/python
cachedir: .pytest_cache
rootdir: /Users/weijie/code/StockyAI
configfile: pytest.ini
plugins: cov-7.1.0, asyncio-1.3.0, anyio-4.13.0
asyncio: mode=Mode.STRICT, debug=False, asyncio_default_fixture_loop_scope=None, asyncio_default_test_loop_scope=function
collected 14 items

tests/integration/test_bot.py::TestHandleStart::test_start_clears_history_and_adds_user PASSED
tests/integration/test_bot.py::TestHandleStart::test_start_onboarded_user_gets_welcome_back PASSED
tests/integration/test_bot.py::TestHandleMessage::test_onboarding_gate PASSED
tests/integration/test_bot.py::TestHandleMessage::test_text_message_routes_to_agent PASSED
tests/integration/test_bot.py::TestHandleMessage::test_voice_message_transcribed PASSED
tests/integration/test_bot.py::TestHandleMessage::test_voice_transcription_failure PASSED
tests/integration/test_bot.py::TestHandleMessage::test_forwarded_message PASSED
tests/integration/test_bot.py::TestHandleCallback::test_approve_callback PASSED
tests/integration/test_bot.py::TestHandleCallback::test_edit_callback PASSED
tests/integration/test_bot.py::TestHandleCallback::test_approve_missing_draft PASSED
tests/integration/test_bot.py::TestHandleCommandTrigger::test_trigger_morning_brief PASSED
tests/integration/test_bot.py::TestHandleCommandTrigger::test_trigger_default_is_morning PASSED
tests/integration/test_bot.py::TestSendResponse::test_normal_response PASSED
tests/integration/test_bot.py::TestSendResponse::test_draft_response_with_keyboard PASSED

============================================================= 14 passed in 0.70s ============================================================
```

3. Defect Log & Fix Traceability (Root Cause Analysis)

Bug ID	Bug Description	Steps to Reproduce	Console/Error Log Snippet	Root Cause Analysis (Why did it break?)	Fix Commit Hash / PR Link
DEF-01	Github Actions Workflow CI Fix	In the github-ci.yml file, change docker tag ${{ vars.ECR_REPOSITORY }}:latest to docker tag ${{ secrets.ECR_REPOSITORY }}:latest \	Log error on github CI	It broke because the correct variable to fetch environment variables are not called, therefore the necessary secrets to run the application is missing	7090fe32edc0f4016ce9bf973cc019b87cc16f56

---

4. Known Issues & Deferred Technical Debt

Bug ID	Description & Impact (Bug or Technical Debt)	Reason for Deferral	GitHub Issue Link
DTD-01	In services/weather.py, the `CITY_COORDS` dictionary only contains 5 Malaysian cities. If a trader inputs a city that isn't on that list, the script silently defaults to Kuala Lumpur's coordinates without	Low priority as the weather has already been proved working correctly. The hardcoded cities are just to showcase the working weather api fetching.	Issue 1

Bug ID	Description & Impact (Bug or Technical Debt)	Reason for Deferral	GitHub Issue Link
	warning, skewing spoilage/rain forecasts for users outside the Klang Valley.		

---

5. Live Demo / UAT Risks

- **Risk 1: Voice Transcription Accuracy.** Voice notes are transcribed locally using the faster-whisper "tiny" model to prioritize speed (~1 second). Heavy background noise or thick, non-standard dialects may result in inaccurate transcripts. Please speak clearly into the microphone.

- **Risk 2: Rate Limiting & Network Latency.** The agent relies entirely on the external api.ilmu.ai/v1 endpoint. Slow conference Wi-Fi or sudden rate limits on the API key will delay the Telegram bot's response time.

- **Risk 3: In-Memory Database Locking.** If the environment variable `SUPABASE_DB_URL` is omitted, the app falls back to a local sqlite file. Simultaneous rapid-fire commands from multiple users could potentially lock the SQLite file, though async mitigates this partially.